

a)

Motional emf:

$$E = BLv = 0$$

$$I = \frac{V_B - E}{R} = \frac{V_B}{R} = 314.3 \text{ A}$$

$$\rightarrow F = BIL = 283.9 \text{ N}$$

b)

$$I = 0$$

$$\rightarrow V_B = E = BLv \rightarrow v = 122.2 \text{ m/s}$$

c)

$$F = 30 \text{ N}$$

$$\rightarrow BIL = 30 \text{ N} \rightarrow I = 33.33 \text{ A}$$

$$E = V_B - IR = 98.33 \text{ V}$$

$$\rightarrow v = 109.3 \text{ m/s}$$

Mechanical power:

$$P_{out} = F \ell$$

Electrical power:

$$P_{in} = V_B I$$

$$\rightarrow \eta = \frac{P_{out}}{P_{in}} = 89.44\%$$

a)

$$BLI = 11 \text{ N} \rightarrow I = 52.38 \text{ A}$$

$$E = V_B - IR = 96.19 \text{ V} = BLv$$

$$v = 458 \text{ m/s}$$

b)

$$I = \frac{E}{BL}$$

$$E = V_B - IR \rightarrow v = \frac{V_B - \frac{F_a}{BL} R}{BL}$$

$$v = 507.70 \text{ m/s}$$

c)

$$v = 285.5 \text{ m/s}$$

d) { Magnetic density
Voltage source

a)

$$A = (2r)l$$

$$\begin{aligned} e(t) &= BA\omega \sin(\omega t) \\ &= 16.46 \sin(105t) \text{ V} \end{aligned}$$

b)

$$i(t) = \frac{e(t)}{R} = 1.372 \sin(105t) \text{ A}$$

$$I_{\text{rms}} = 0.970 \text{ A}$$

c)

$$\tau(t) = BA i(t) \sin \omega t$$

$$= 0.215 \sin^2(105t) \text{ N.m}$$

It opposes the mechanical rotation.

a)

$$n_s = \frac{120f}{p}$$

Poles	Speed (rpm)
2	1500
4	750
6	500
8	375
10	300
12	250
14	214.3

b)

$$P_n \propto f$$

$$P_e \propto f^2$$

$$\frac{P_{n,25}}{P_{n,60}} = 41.7\%$$

$$\frac{P_{e,25}}{P_{e,60}} = 17.4\%$$

c)

Light flicker decreases as frequency increases.

At 25 Hz, Light flicker is noticeable.

d)

$$P_{elec} = I_{rms}^2 R = 11.3 \text{ W}$$

e)

$$P_{mech} = T_{avg} \omega = \frac{T_{max}}{2} \omega = 11.3 \text{ W}$$

P_{elec} and P_{mech} are equal as expected from conservation of energy